



ArguLens: An Open-Source System for Automated Essay Scoring and Label-Aware Feedback Generation

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Abstract

Most automated essay scoring (AES) systems output a single holistic score without interpretable evidence and rely on closed APIs that introduce data privacy and cost barriers. We present **ArguLens**, an open-source, locally deployable system that decomposes AES into three decoupled components: a discourse-move classifier (Qwen2.5-7B-Instruct fine-tuned with LoRA on PERSUADE 2.0), a grade-independent LightGBM scorer over 31 linguistic and discourse features, and a label-aware feedback generator served through vLLM with a Qwen2.5-14B-Instruct backbone. A Gradio web UI exposes pluggable inference backends and supports single-essay and batch scoring with downloadable per-essay breakdowns. On an essay-disjoint PERSUADE 2.0 test split, the logit-probe classifier achieves 82.6% accuracy and 0.727 macro-F1; under prompt-grouped 5-fold cross-validation the scorer reaches a mean QWK of 0.813 under an oracle discourse-feature protocol, and an ablation shows that adding gold discourse annotations yields an increment of +0.055 QWK over the lexical+syntactic configuration (paired t -test, $p = 0.010$). This is a component-level diagnostic rather than an end-to-end classifier-to-scorer result. The feedback generator ships with a structured evaluation protocol; its human-rater study is left to future work. The system is released under Apache 2.0 at https://github.com/wrwbs/AI_AWE.

1 Introduction

Automated essay scoring (AES) has evolved from linear regression on surface features (Page, 1966) to deep neural architectures (Taghipour and Ng, 2016; Mayfield and Black, 2020) and, most recently, to LLM-based holistic scoring (Beigman Klebanov and Madnani, 2020; Tate et al., 2024). Despite this progress, a fundamental

tension persists between interpretability and performance. Classic feature-based methods offer transparent scoring but cap ceiling accuracy; neural models improve accuracy at the cost of opacity, and LLM prompting requires closed APIs that restrict local deployment and data privacy.

Three specific gaps remain for research and assistive deployment in educational contexts. First, end-to-end models output a single score without interpretable linguistic or rhetorical evidence, making it difficult for instructors to understand how a grade was reached. Second, reliance on commercial APIs introduces cost, latency, and data-privacy barriers that limit adoption in resource-constrained settings. Third, most AES systems stop at a holistic score; fine-grained, label-aware revision suggestions that could help learners improve their writing are rare.

We address these gaps with **ArguLens**, a locally deployable, Apache 2.0-licensed system whose design and evaluation are presented in this paper. The system:

- decomposes scoring into discourse-move classification, feature extraction, LightGBM scoring, and label-aware feedback;
- supports local deployment via vLLM with tensor parallelism and a 4-bit HuggingFace fallback, while retaining an optional API backend;
- ships reproducible artifacts including the scorer model, scaler, LoRA adapter configuration in the repository, and weights as a versioned Release asset with checksums;
- exposes a config-driven stack with zero hard-coded paths, environment-variable precedence, and an offline test suite.

We evaluate the classifier and scorer on the PERSUADE 2.0 corpus (Crossley et al., 2024) of

US middle-school argumentative essays graded 1 to 6. The discourse-move classifier is assessed through sentence-level accuracy and macro-F1; the scorer is evaluated through quadratic weighted kappa (QWK), accuracy, and macro-F1 under prompt-grouped cross-validation. The feedback component is evaluated here at the level of its documented generation workflow and repository contract; human-rater results are explicitly outside the scope of this release.

2 Related Work

AES Paradigms. Automated writing evaluation has a long history of balancing predictive accuracy, interpretability, and instructional use (Beigman Klebanov and Madnani, 2020). Classic feature engineering (Kyle and Crossley, 2015; Lu, 2010) exposes linguistic evidence, whereas neural AES models (Taghipour and Ng, 2016; Mayfield and Black, 2020) can learn richer representations at the cost of transparency. Recent work also shows that general-purpose AI can provide useful holistic scoring, while raising questions about rubric alignment and validity (Tate et al., 2024). Our grade-independent LightGBM scorer retains an explicit feature space while the LoRA classifier adds rhetorical structure to it.

Argument Structure and Feedback. Argument mining in persuasive and student essays has modeled both component structure and essay quality (Persing and Ng, 2010; Stab and Gurevych, 2017). Recent work further reports that combining argument-segment and cohesion features improves automatic feedback-related scoring (Ding et al., 2024). Recent benchmarking makes the granularity issue explicit: EssayJudge evaluates AES at lexical, sentence, and discourse levels and still finds substantial gaps at the discourse level (Su et al., 2025). We adopt the four PER-SUADE 2.0 labels (claim, data, counterclaim, rebuttal) and fine-tune a Qwen2.5-7B LoRA, avoiding full-parameter updates while preserving the base model’s language understanding.

Feedback Evaluation and Fairness. Feedback quality is multidimensional rather than reducible to a single automatic score. LLM-Rubric demonstrates the value of explicit dimensions and calibration against human judgments when evaluating generated text (Hashemi et al., 2024). More recent writing-feedback evaluation finds that models

may produce specific comments while missing the most important problem or misjudging whether criticism is appropriate (Rashkin et al., 2025). Accordingly, this release defines relevance, actionability, and tone as evaluation dimensions but does not report human ratings. Fairness is also not implied by aggregate AES accuracy: subgroup performance can vary with demographic and learner characteristics (Schaller et al., 2024). We therefore treat subgroup analysis as a required future study rather than making a fairness claim from the aggregate metrics reported here.

Open Educational AI. Our release follows model-reporting practice by documenting the model, training context, limitations, and asset locations (Mitchell et al., 2019). Small artifacts are kept in the repository, while the larger adapter is distributed as a versioned Release asset with a checksum. This is a release design choice, not an experimental finding.

3 System Architecture

Figure 1 illustrates the end-to-end pipeline. The Gradio frontend (`essay_score/app_gradio.py`) orchestrates three independent services whose responsibilities are summarised in Table 1. The move classifier runs through pluggable HF or vLLM backends and emits sentence labels; the feature extractor combines TAALED lexical indices, QuanSyn dependency metrics, and discourse counts into a 31-feature vector; the scorer outputs a score, confidence, and flags; and the feedback generator produces structured feedback through vLLM or an OpenAI-compatible API.

Table 1: Core modules and their responsibilities.

Module	Path	Responsibility
Discourse-Move Classifier	<code>essay_score/infer</code>	Sentence-level 4-way classification (LoRA on Qwen2.5-7B)
Scorer	<code>essay_score/scoring_pipeline</code>	31-feature LightGBM multiclass (scores 1–6)
Feedback Generator	<code>essay_score/feedback</code>	Label-aware English feedback via Qwen2.5-14B (vLLM)
Frontend	<code>essay_score/app_gradio.py</code>	Gradio UI, batch processing, and ZIP export
Training Code	<code>qwen_move_classifier</code>	LoRA SFT with LLaMA-Factory, data prep



Figure 1: Project workflow for linguistically informed writing assessment. The upper lane summarizes offline training and deployment; the lower lane shows online assessment, scoring, grounded feedback, and the personalized report.

3.1 Configuration and Reproducibility

All runtime settings follow a strict precedence: environment variables override `config.yaml`, which in turn overrides default HuggingFace model IDs. The configuration loader (`essay_score/config.py`) resolves relative paths against the repository root. A `model_checksums.sha256` manifest enables `sha256sum -c` verification of the scorer, scaler, adapter config, tokenizer, and LoRA weights, ensuring that downstream users can verify asset integrity.

4 Discourse-Move Classifier

4.1 Task Formulation

Given an essay split into N sentences $\{s_1, \dots, s_N\}$, the classifier predicts a label $y_i \in \{\text{claim}, \text{data}, \text{counterclaim}, \text{rebuttal}\}$ for each sentence. It operates at sentence level with local context derived from the preceding sentence and the essay opening.

4.2 Base Model and LoRA

The classifier uses the Apache-2.0-licensed Qwen/Qwen2.5-7B-Instruct model (Qwen et al., 2025) as its base. A LoRA adapter (Hu et al., 2021) with rank $r = 32$, $\alpha = 64$, and dropout 0.05 is applied to all linear projection

modules (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`). Training was conducted with LLaMA-Factory (Zheng et al., 2024) on PERSUADE 2.0 using 4-GPU DDP (`torchrun`) and BF16 precision, with a peak learning rate of 1×10^{-4} under a cosine schedule (warmup ratio 0.05), weight decay 0.01, a cutoff length of 2,048 tokens, two epochs, and an effective batch size of 32 (4 GPUs $\times$ batch 1 $\times$ gradient accumulation 8). The repository ships the adapter configuration files; the weight file (`adapter_model.safetensors`, 309 MB) is published as a GitHub Release asset tagged `v0.1.0`.

4.3 Training Data and Model Selection

PERSUADE 2.0 does not prescribe an official split, so we created an essay-level split with seed 42: 22,605 essays for training, 2,825 for validation, and 2,826 for testing, with zero essay overlap between all pairs of splits (verified programmatically). Because the natural label distribution is heavily skewed (data and claim dominate), the training set is class-balanced by resampling to 25,000 sentences per label (100,000 examples in total). A small balanced validation set (2,000 sentences, 500 per label) monitors training, and the checkpoint with the lowest validation loss is selected for evaluation; all reported test results use

the natural-distribution test set of 12,000 sentences from 2,730 essays.

4.4 Logit-Based Classification

Rather than generating free text, the classifier uses a discriminative logit probe for deterministic inference. For each target sentence, a chat-template prompt is built with a system definition of the four move types encoded as single-token codes (A, B, C, D). A forward pass through the base model with the attached LoRA produces logits at the last non-padding position. The four logits corresponding to the code tokens are extracted and passed through softmax; the $\arg \max$ is the predicted label and the maximum softmax probability is the per-sentence confidence score. The HF backend chunks prompts using a configurable sentence batch size, while the vLLM backend sends a bounded request to a persistent service with one-token decoding and the four allowed label tokens.

4.5 Inference Backends

Three backends are available. The `hf` backend loads the base model and adapter in-process using `transformers` and `peft`. The `vllm` backend runs a persistent FastAPI service that loads the base model once and attaches the LoRA via `LoRARequest` (Kwon et al., 2023). Its `/health` endpoint reports the adapter rank, tensor parallelism size, and GPU count. The frontend falls back to `hf` if the vLLM service is not ready. The `api` backend is reserved for the feedback generator.

5 Grade-Independent LightGBM Scorer

5.1 Feature Set

All features are computed without learner grade information to prevent direct target leakage. This design does not establish fairness, because linguistic features can still correlate with demographic or curricular variables. Extraction uses a single shared spaCy dependency parse for both lexical and syntactic metrics.

Lexical Features (TAALED (Kyle and Crossley, 2015)). Sixteen lexical features comprise twelve lexical-diversity variants across AW, content-word, and function-word classes, together with `word_count`, `awl`, `basic_nfunction_tokens`, and `lexical_density_tokens`.

Syntactic Features (QuanSyn (Yang and Liu, 2025)). Eight dependency-based features are retained from QuanSyn: `mdd`, `ndd`, `mhd`, `mtdl`, `vk`, `mtw`, `hi`, and `mrd`.

Discourse Features (deployment and evaluation protocols). The deployment pipeline derives seven features from the Qwen2.5-7B LoRA move-classifier output: `n_sent`, `n_claim`, `n_data`, `pct_data`, `pct_counterclaim`, `pct_rebuttal`, and `has_counter_rebut` (binary, true only when both a counterclaim and a rebuttal are present). The scorer feature table used for the reported cross-validation instead contains the corresponding PERSUADE gold discourse annotations. We therefore call the reported scorer results an oracle component evaluation and do not present them as an end-to-end classifier benchmark.

Handling Missing Dependencies. If the `quansyn` package is unavailable, the scorer raises a `FeatureExtractionError` by default. A configuration flag (`allow_missing_dependency_features`) fills missing values with zeros for exploratory runs only.

5.2 Model Training

The scorer uses LightGBM (Ke et al., 2017) with the `multiclass` objective over six ordinal classes. Hyperparameters include `num_leaves=63`, `learning_rate=0.03`, `n_estimators=1000`, `min_child_samples=15`, `subsample=0.8`, `colsample_bytree=0.8`, and `reg_lambda=0.1`. Training uses a seed of 42; each cross-validation fold increments the model seed by its fold index. Training data are the PERSUADE 2.0 grade-independent feature table, with prompt groups held out during cross-validation. The trained model (`lightgbm_multiclass.txt`, 33 MB) and the feature scaler (`scaler.json`) are committed to the repository.

At inference, raw feature values are z-score normalized and passed to the LightGBM model, which outputs a six-class probability vector. The expected value is rounded to the nearest integer in 1–6 to produce the final score. The same expected-value decoding rule is used by

cross-validation, ablation, and production inference. The reported cross-validation uses the `gold_annotations` discourse-feature source; production replaces those fields with predicted move labels from the classifier. The system flags essays with `low_confidence` when the maximum class probability falls below 0.5 and marks them as `borderline` when the runner-up probability exceeds 0.30 and is adjacent to the predicted class; both thresholds are fixed heuristics rather than tuned values.

6 Label-Aware Feedback Generation

6.1 Prompt Design

Feedback is generated on every analysis. The prompt template (`feedback/prompt_essay_v2.json`) receives the essay text, the essay prompt name, the predicted score, confidence, and flags, together with a curated subset of 14 of the 31 features—four lexical (e.g., AWL, MTLD), five syntactic (MDD, MHD, MRD, MTDL, head-initial ratio), three discourse (claim and evidence counts, counterclaim/rebuttal indicator), and two length features. Each feature is accompanied by a band label (Low, Medium, or High) derived from approximate corpus-level PERSUADE 2.0 tertiles; for example, an AWL below 4.22 is labelled Low, between 4.22 and 4.54 Medium, and above 4.54 High, while MDD, MHD, MRD, and MTDL use lower-is-better scoring. These bands are used only for prompting the feedback model and never enter the scorer, so they cannot leak score information. The system prompt enforces constructive, English-only output, forbids inventing metrics, requires quoting real essay text, and instructs the model to recommend human review whenever the low-confidence or borderline flag is set. The output contract fixes exactly five sections: overall assessment, lexical dimension, syntactic dimension, discourse/argumentation dimension, and the top one or two actionable improvements.

6.2 Backends

Three backends are available. The `vllm` backend (default) runs a persistent OpenAI-compatible server (port 8000, TP=2 for the 14B model). The `hf` backend uses 4-bit `BitsAndBytesConfig` with SDPA on a single GPU. The `api` backend accepts any OpenAI-compatible endpoint; the API key is read from the UI or environment at runtime

and is never stored.

7 Frontend, Batch Processing, and Export

The Gradio 6.x interface (`essay_score/app-gradio.py`) provides single-essay analysis, batch upload with summary table, and ZIP export. Each per-essay folder in the archive contains `essay.txt`, `score.json`, `linguistic_features.json`, `discourse_features.json`, `feedback.md`, `result.json`, and `sentence_labels.csv`. The root contains `summary.csv` and `manifest.json`.

8 Experiments and Evaluation

We evaluate the classifier and scorer using distinct protocols because the available artifacts are at different granularities. The classifier uses the essay-disjoint sentence-level test split described in Section 4.3; the scorer uses prompt-grouped cross-validation on the grade-independent feature table. We do not combine these protocols into a single end-to-end score. Because the grouping variable is the essay prompt, each scorer fold withholds complete prompts rather than randomly splitting essays; this is a cross-prompt estimate, not an iid random-split estimate. The released feature table does not include the raw essay text needed to regenerate predicted discourse features for all scorer essays, so the reported scorer numbers use gold discourse annotations and are explicitly an oracle component evaluation.

8.1 Experimental Setup

Dataset. PERSUADE 2.0 (Crossley et al., 2024) contains more than 25,000 argumentative essays written by US students in grades 6–12 across 15 prompts; each essay is scored holistically on a 1–6 rubric. The corpus is available under a non-commercial Creative Commons license from [PERSUADE 2.0 repository](#).

Metrics. For the discourse-move classifier, we report sentence-level accuracy, macro-F1, and weighted-F1. For the LightGBM scorer, we report quadratic weighted kappa (QWK) between predicted and human-assigned scores and classification accuracy, both with standard deviations over the five prompt-grouped folds. A protocol for evaluating the feedback generator on relevance, actionability, and tone is defined in the repository;

Table 2: Discourse-move classifier results on the PERSUADE 2.0 test set (12,000 sentences from 2,730 essays).

Label	Precision	Recall	F1	Support
claim	0.776	0.835	0.805	4,209
data	0.903	0.825	0.862	7,186
counterclaim	0.501	0.761	0.604	343
rebuttal	0.533	0.794	0.638	262

Overall: accuracy 82.58%, macro-F1 0.727, weighted-F1 0.830

human-annotator results are out of scope for this report. Inference latency is reported as median and 95th percentile (p95) over 100 independent runs on the reference hardware (1×RTX 5000 Ada, 32 GB).

Ablation design. For the scorer, we ablate the 31-feature model against two reduced feature sets: (1) the sixteen lexical features only (no discourse or syntactic features); (2) lexical plus eight syntactic features (no discourse features). The delta between the full model and configuration (2) measures the marginal value of the seven gold discourse features. It is an oracle diagnostic, not a measurement of the downstream classifier’s gain. No additional baseline was run in this release; comparisons against ordinal logistic regression, zero-shot or generative prompts, and fine-tuned encoder classifiers remain future work rather than unreported results.

8.2 Classifier Evaluation

Table 2 reports the discourse-move classifier performance on the PERSUADE 2.0 test set, and Table 3 gives the corresponding confusion matrix. The `data` class achieves the highest F1 (0.862), followed by `claim` (0.805). The two classes with the largest error mass are `claim` and `data`: confusion between them accounts for 1,528 of the 2,091 misclassified sentences (73%), reflecting the genuine difficulty of separating evidence sentences from claims that embed evidence. The minority classes `counterclaim` (F1 0.604) and `rebuttal` (F1 0.638) show low precision (0.501 and 0.533) but comparatively high recall (0.761 and 0.794), a pattern consistent with the class-balanced resampling used during training. Support counts are reported so that the macro-F1 is not misread as performance on a balanced test set.

8.3 Scorer Evaluation

Table 4 reports the LightGBM scorer performance. Fold-level QWK for the full model ranges

Table 3: Confusion matrix on the PERSUADE 2.0 test set (rows: gold label; columns: prediction).

	claim	data	counter.	rebut.
claim	3,515	564	81	49
data	964	5,925	171	126
counterclaim	25	50	261	7
rebuttal	23	23	8	208

Table 4: Scorer performance: QWK and accuracy (mean and standard deviation over prompt-grouped 5-fold cross-validation on the 25,996-essay grade-independent feature table).

Model	QWK ↑	Acc. (%) ↑
<i>Our approach</i>		
LightGBM (all 31 features)	0.813 (0.048)	62.73 (2.44)
<i>Ablations</i>		
LightGBM (lexical only, 16 feats)	0.750 (0.071)	57.57 (2.16)
LightGBM (lexical + syntactic, 24 feats)	0.759 (0.069)	58.69 (2.32)

from 0.754 to 0.872, indicating heterogeneous difficulty across withheld prompts.

Ablation: oracle discourse features. Adding gold discourse features yields +0.055 QWK over lexical+syntactic features alone (0.813 vs. 0.759) and +0.064 over lexical-only features (0.813 vs. 0.750). Because the five folds are prompt-aligned across configurations, we can pair them: all five per-fold differences are positive, with a mean paired difference of +0.057 (full vs. lexical+syntactic; paired $t(4) = 4.64$, $p = 0.010$) and +0.064 (full vs. lexical-only; paired $t(4) = 5.04$, $p = 0.007$). A Wilcoxon signed-rank test cannot fall below $p = 0.0625$ with five pairs, so we report the t -test and note that these results should be read as a descriptive, component-level diagnostic rather than conclusive evidence about predicted classifier features in deployment.

8.4 Feedback Generation and Evaluation Protocol

The feedback generator produces label-aware, constructively framed revision suggestions from the predicted discourse-move labels and feature bands (see Section 6). A structured evaluation protocol of feedback quality (relevance, actionability, and tone) is defined in the repository; human-annotator results are left for future work and are not part of this report.

8.5 Latency Benchmarks

Table 5 reports component inference latency measured on a single RTX 5000 Ada (32 GB VRAM)

over 100 independent runs; it is not a full-pipeline latency measurement.

Table 5: Inference latency by component and backend. Hardware: 1×RTX 5000 Ada (32 GB); 100 runs per component.

Component	Backend	p50 (s)	p95 (s)
Classifier (21-sentence reference essay)	HF (BF16)	1.30	1.32
Scorer (one essay)	CPU (32 threads)	0.006	0.009

GPU memory usage. The classifier (HF backend, BF16, Qwen2.5-7B + LoRA) uses a peak of 16.7 GB allocated GPU memory (18.5 GB reserved) for a single-essay inference with 21 sentences. We do not report a memory or latency number for the Qwen2.5-14B feedback backend because it was not benchmarked in this release; consequently, the reported latency is not an end-to-end full-pipeline latency.

8.6 Summary of Experimental Findings

The experiments yield four main findings. First, the logit-probe classifier with Qwen2.5-7B + LoRA achieves 82.58% accuracy and a macro-F1 of 0.727 on the 12,000-sentence PERSUADE 2.0 test set, with claim–data confusion accounting for 73% of the errors. Second, the full 31-feature LightGBM scorer achieves a mean QWK of 0.813 over 5-fold CV (fold range 0.754–0.872), with gold discourse features contributing an increment of +0.055 QWK over the lexical+syntactic configuration in the oracle component evaluation (paired *t*-test, $p = 0.010$). Third, the scorer runs at sub-10-ms scorer-only latency on CPU, while the classifier completes a 21-sentence essay in approximately 1.3 seconds on a single RTX 5000 Ada with a peak GPU memory of 16.7 GB. Fourth, the test suite passes 22/22 offline tests, supporting the reproducibility claims of the release.

9 Open-Source Release Strategy

10 Testing and Quality Assurance

The `tests/` suite validates eight aspects of the system, as listed in Table 7. The suite is intentionally offline: the vLLM API contract test uses a fake engine rather than requiring a live model service.

Table 6: Asset distribution policy.

Asset	Location	Reason
Source code, documentation, license	Git repository	Version control
Adapter config + tokenizer	Git repository (1.2 MB)	Small, user IP
LightGBM model + scaler	Git repository (33 MB)	Offline reproducibility
LoRA weights	GitHub Release (309 MB)	Avoids Git LFS bandwidth
Base models (Qwen2.5-7B, 14B)	HuggingFace (Apache 2.0)	Hub License restriction
PERSUADE 2.0 corpus	Upstream non-commercial license, repository	Non-commercial research use

Table 7: Test coverage summary.

Test module	Covers	Pass rate
Batch outputs	ZIP structure, manifest, CSV	3/3
Classifier prompt	Prompt template, label parsing, shared context contract	4/4
Feedback bands	Band thresholds	4/4
Frozen assets	Adapter config, tokenizer, SHA-256	1/1
Gradio contract	I/O schema (single + batch), branding	4/4
Model configuration	Config precedence, path resolution	2/2
Scorer contract	Feature order, scaler, flags, score decoder	3/3
vLLM API	API contract (fake engine)	1/1
Total		22/22

All 22 tests pass without a GPU or network connection.

11 Conclusion

This paper presented ArguLens, an open-source modular system for automated essay scoring and label-aware feedback. The central contribution is a decoupled pipeline that separates discourse classification, feature-based scoring, and feedback generation, allowing each component to be developed, tested, and replaced independently.

Our experiments show that the logit-probe classifier achieves 82.58% accuracy and a macro-F1 of 0.727 on the PERSUADE 2.0 test set. The LightGBM scorer with all 31 features obtains a QWK of 0.813 over 5-fold CV. In the oracle component evaluation, adding gold discourse features contributes an increment of +0.055 QWK over lexical and syntactic features alone (paired *t*-test, $p = 0.010$); this should not be interpreted as the gain from predicted classifier features. The feedback generator’s evaluation protocol is defined in the repository; a human-annotator study of feedback actionability is left to future work. The reported component benchmarks are approximately 1.3 seconds for classification and 6–9 ms for scoring; the feedback backend and therefore full-pipeline latency were not benchmarked.

Extending the pipeline beyond English argu-

mentative essays in a single educational context — using an XLM-RoBERTa backbone for discourse classification and multilingual LLMs for feedback — is a natural next step, as is a larger-scale human evaluation of feedback actionability across diverse learner populations. The code and release metadata are provided under Apache 2.0; third-party models and data retain their upstream licenses. The evaluation protocol and model artifacts are released to facilitate community-driven extensions.

12 Responsible Use and Limitations

The system is trained exclusively on US middle-school argumentative essays from PERSUADE 2.0. Its performance should not be expected to generalise to other genres, age groups, or languages without domain-specific fine-tuning. The corpus reflects specific demographic and curricular contexts, and scores may carry demographic biases. This release does not report gender, race/ethnicity, or other subgroup metrics; a future study should use the corpus metadata, report subgroup support and uncertainty, and evaluate both score accuracy and feedback quality before any educational deployment. Pending such analysis, the system is intended for research and assistive classroom use only, not for high-stakes automated decisions.

The classifier is evaluated on completed essays and is not tested on partial drafts. This matters for formative use: argument-mining models can be less robust when context is incomplete, even when they perform well on completed essays (Schaller et al., 2025). The scorer also uses a single holistic 1–6 target and does not model rater-level variation, a limitation when interpreting borderline cases (Gaudeau, 2025).

A further methodological caveat is that the two released evaluation protocols are not an end-to-end evaluation. The classifier is evaluated on an essay-disjoint sentence-level split, whereas the reported scorer cross-validation uses a feature table whose discourse fields are PERSUADE gold annotations. Thus, the +0.055 QWK result is an oracle component diagnostic, not a claim about predicted classifier features. A prompt-held-out end-to-end rerun requires the licensed raw essay text and will be the next evaluation step; comparisons against ordinal logistic regression, zero-shot or generative prompting baselines, and fine-tuned encoder classifiers remain future work.

The classifier benchmark used 16.7 GB allocated and 18.5 GB reserved VRAM on one RTX 5000 Ada. The scorer itself runs on CPU, while the feedback backend can call an OpenAI-compatible service; resource requirements for the local 14B feedback deployment were not measured. The LoRA weights must be downloaded separately and verified against the SHA-256 checksum before first use.

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A Reproducibility Checklist

- **Code:** https://github.com/wwrwbs/AI_AWE (Apache 2.0)
- **LoRA weights:** https://github.com/wwrwbs/AI_AWE/releases/tag/v0.1.0
- **Models:** Qwen2.5-7B-Instruct and Qwen2.5-14B-Instruct (HuggingFace Hub, Apache 2.0)
- **Data:** PERSUADE 2.0 (upstream non-commercial license; [data repository](#))
- **Environment:** frontend, vLLM, and training lockfiles (the three version-pinned requirement files)
- **Checksums:** `model_checksums.sha256`
- **Tests:** `pytest -q` (offline)
- **Launch:** `run_all_services.sh` (auto-detects virtual environments, GPUs, and ports)
- **LoRA training config:** `qwen_move_classifier/configs/qwen25_7b_move_lora_sft.yaml`

- **Split statistics:** `qwen_move_classifier`
/data/dataset_summary.json (essay-level split with seed 42; zero overlap between splits)

B Full Feature List

Table 8 enumerates all 31 features used by the scorer, grouped by family. Lexical-diversity variants are computed over all words (`_aw`), content words (`_cw`), and function words (`_fw`).

Table 8: The 31 scorer features.

Feature	Description
<i>Lexical (16; TAALED and basic counts)</i>	
<code>word_count</code>	Total word count
<code>awl</code>	Average word length
<code>basic_nfunction_tokens</code>	Function-word token count
<code>lexical_density_tokens</code>	Lexical density (content tokens / total tokens)
<code>mstr50_*</code>	Moving-average segmental TTR, window 50
<code>mtld_original_*</code>	Measure of Textual Lexical Diversity
<code>mtld_ma_bi_*</code>	Bilinear moving-average MTLD variant
<code>hdd42_*</code>	HD-D hypergeometric diversity, 42 draws
<code>simple_ttr_fw</code>	Simple type-token ratio (function words)
<code>root_ttr_fw</code>	Root TTR / Guiraud’s index (function words)
<i>Syntactic (8; QuanSyn dependency metrics)</i>	
<code>mdd</code>	Mean dependency distance
<code>ndd</code>	Normalized dependency distance
<code>mhd</code>	Mean hierarchical distance
<code>mtdl</code>	Mean total dependency length
<code>vk</code>	Valency k index
<code>mtw</code>	Mean tree width
<code>hi</code>	Head-initial ratio
<code>mrd</code>	Mean root distance
<i>Discourse (7; predicted in deployment, gold in oracle evaluation)</i>	
<code>n_sent</code>	Sentence count
<code>n_claim / n_data</code>	Claim / evidence sentence counts
<code>pct_data</code>	Share of data sentences (%)
<code>pct_counterclaim</code>	Share of counterclaim sentences (%)
<code>pct_rebuttal</code>	Share of rebuttal sentences (%)
<code>has_counter_rebut</code>	Binary: essay contains both counterclaim and rebuttal

* computed over all words (`_aw`), content words (`_cw`), and function words (`_fw`).